Practice Problem 5 Guide Questions

1. **Why is it beneficial to split the program into two separate functions, get_reverse() and is_palindrome(), instead of writing all the code inside the main() function?**
   It is easier to work on many small tasks rather than one big task. The main function would become messy if it handled too many tasks directly. These functions are also easy to reuse if we need to call them multiple times.

2. **Suppose num contains the value 1234. How can the remainder operator (%) be used to obtain the last digit of num? How can integer division be used to remove the last digit?**
   `1234 % 10 = 4`
   `1234 / 10 = 123`

3. **When reversing a number, the digits are processed one at a time starting from the last digit. Suppose reverse currently contains 43 and the next digit is 2. What arithmetic operation can be used to change 43 into 432?**
   `reverse = reverse * 10 + digit`

4. **Suppose one of the test cases is the number 0. What would the program output for this test case? Explain why this result makes sense based on how the loop in get_reverse() works.**
   It correctly outputs 0. The loop does not iterate because 0 has been reached. The reverse number remains at its initial value of 0.

5. **Suppose one of the test cases is the number 1000. The reverse of this number would be 0001, which is stored simply as 1. What output should the program produce in this case, and why is this the expected behavior?**
   The program determines that 1000 does not equal 1, and correctly determines that 1000 is not a palindrome. In general, any input with trailing zeros will be correctly identified as not a palindrome. A trailing zero would have to be paired with a leading zero for the number to be a palindrome, but leading zeros are impossible for our integer input.

6. **Suppose one of the test cases is a very large number close to the maximum value an int can hold in C (for example, 2,147,483,647). What potential problems might occur when reversing this number, and why? What data type could be used instead to help avoid overflow?**
   Overflow will occur if the expected reversed number exceeds the integer limit. This will specifically happen when the reversed number is being shifted into the billions place. We can use a long for extra storage. Arrays or strings could give effectively more storage, but these haven't been taught yet.

7. **The for loop in main() uses the variable i as its loop control variable. Can the variable i be accessed outside the loop after it finishes executing? Explain why or why not, based on block scope rules in C.**
   It cannot be accessed outside the loop. Variables declared in a for-loop init-clause are scoped to that loop. Only the body of the loop and the other loop header expressions can access these variables.